

	Olist E-Commerce Analytics
Brief Ref.	QL-OL-001
Issued By	Analytics Division — Quantify Labs
Issued To	Data Analyst & Data Engineer
Issue Date	23 May 2026
Deadline	30 May 2026 — 17:00 UTC
Priority	HIGH
Review By	Team Lead — Quantify Labs

1. PROJECT CONTEXT

Quantify Labs has been engaged by Olist, a Brazilian e-commerce marketplace operator, to deliver a structured analytical layer on top of their PostgreSQL transactional database. The database captures the full order lifecycle — from customer and seller registration through to payment, delivery, and post-purchase review.

The client has expressed a need to move from ad-hoc reporting to a governed, scheduled, and reproducible analytical framework. Your assignment covers both the analytical SQL layer and the engineering pipeline that keeps it current.

2. OBJECTIVES

By the deadline, you are expected to have produced verified, reviewed, and documented outputs across all three objectives below.

#	Objective	Acceptance Criteria
1	Analytical SQL Layer	17 analytical queries authored, tested, and returning correct results against the live schema. Queries must span both simple and advanced SQL features.
2	Materialized View Layer	4 materialized views created with UNIQUE indexes. Each must support CONCURRENTLY refresh without locking. Views must be tested and validated.
3	Airflow Pipeline	4 DAGs implemented: daily refresh, weekly aggregates, monthly reports, and data quality checks. Each task must have retry logic and documented failure behaviour.

3. DELIVERABLES

The following must be submitted to the internal review folder by the deadline:

Deliverable	Format	Status
All 17 analytical SQL queries	.sql file	Pending
4 Materialized view DDL scripts	.sql file with CREATE and INDEX statements	Pending
4 Airflow DAG files	Python .py — PEP8 compliant	Pending
Aggregate sink table DDL	.sql file with CREATE TABLE statements	Pending
This task brief — signed off	Return annotated copy to team lead	Pending

4. TASK BREAKDOWN & SEQUENCE

Complete tasks in the order listed. Do not move to the next task without confirming the previous one passes review.

1. Task 001-A — Write and validate simple SQL queries (Q1–Q8). Submit for review.
2. Task 001-B — Write and validate advanced SQL queries (Q9–Q17). Submit for review.
3. Task 001-C — Author and test all 4 materialized view DDL scripts. Confirm CONCURRENTLY refresh works.
4. Task 001-D — Build Airflow DAGs: daily, weekly, monthly, data quality. Unit test each task in isolation.
5. Task 001-E — Author aggregate sink table DDL and confirm Airflow write tasks resolve without errors.
6. Task 001-F — Final documentation pass. Submit full deliverable package to team lead.

5. QUANTIFY LABS SQL STANDARDS

All SQL submitted must comply with the following internal standards. Non-compliant submissions will be returned without review.

- All queries must be written in PostgreSQL-compatible SQL.
- CTEs must be used in preference to nested subqueries where clarity demands it.
- All column aliases must be in snake_case and descriptive — no single-letter aliases except in self-joins.
- NULLIF must be used wherever division is performed to prevent zero-division errors.
- COALESCE must handle nullable foreign key joins, particularly product category lookups.
- Window functions must include an explicit ORDER BY within the OVER clause.
- Every query must include a one-line comment block stating its business question.
- Airflow DAGs must include default_args with retries ≥ 2 and retry_delay.

- All DAG SQL must use ON CONFLICT DO UPDATE for idempotent inserts.

6. REVIEW & SIGN-OFF PROCESS

Quantify Labs operates a two-stage review on all client deliverables.

Stage	Description	Reviewer
Stage 1 — Peer	Logic check, standards compliance, output spot-check.	Senior Analyst, Quantify Labs
Stage 2 — Lead	Business question alignment, client presentation readiness.	Team Lead, Data & Analytics Division

Raise blockers in the #olist-project Slack channel. Do not wait until the deadline to surface issues.

7. ACKNOWLEDGEMENT

By commencing work on this brief, you confirm you have read, understood, and accepted the objectives, standards, and deadline above.

Issued By — Quantify Labs	Received By — Analyst / Engineer
Signature: Quantify labs CEO Date: 23 May 2026	Signature: Peter Mchikho Date: _____